



Minimal-Dose Resistance Training for Improving Muscle Mass, Strength, and Function: A Narrative Review of Current Evidence and Practical Considerations

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Abstract

Resistance training (RT) is the only non-pharmacological intervention known to consistently improve, and therefore offset age-related declines in, skeletal muscle mass, strength, and power. RT is also associated with various health benefits that are underappreciated compared with the perceived benefits of aerobic-based exercise. For example, RT participation is associated with reduced all-cause and cancer-related mortality and reduced incidence of cardiovascular disease, hypertension, and symptoms of both anxiety and depression. Despite these benefits, participation in RT remains low, likely due to numerous factors including time constraints, a high-perceived difficulty, and limited access to facilities and equipment. Identification of RT strategies that limit barriers to participation may increase engagement in RT and subsequently improve population health outcomes. Across the lifespan, declines in strength and power occur up to eight times faster than the loss of muscle mass, and are more strongly associated with functional impairments and risks of morbidity and mortality. Strategies to maximise healthspan should therefore arguably focus more on improving or maintaining muscle strength and power than on increasing muscle mass per se. Accumulating evidence suggests that minimal doses of RT, characterised by lower session volumes than in traditional RT guidelines, together with either (1) higher training intensities/loads performed at lower frequencies (i.e. low-volume, high-load RT) or (2) lower training intensities/loads performed at higher frequencies and with minimal-to-no equipment (i.e. resistance ‘exercise snacking’), can improve strength and functional ability in younger and older adults. Such minimal-dose approaches to RT have the potential to minimise various barriers to participation, and may have positive implications for the feasibility and scalability of RT. In addition, brief but frequent minimal-dose RT approaches (i.e. resistance ‘exercise snacking’) may provide additional benefits for interrupting sedentary behaviour patterns associated with increased cardiometabolic risk. Compared to traditional approaches, minimal-dose RT may also limit negative affective responses, such as increased discomfort and lowered enjoyment, both of which are associated with higher training volumes and may negatively influence exercise adherence. A number of practical factors, including the selection of exercises that target major muscle groups and challenge both balance and the stabilising musculature, may influence the effectiveness of minimal-dose RT on outcomes such as improved independence and quality-of-life in older adults. This narrative review aims to summarise the evidence for minimal-dose RT as a strategy for preserving muscle strength and functional ability across the lifespan, and to discuss practical models and considerations for the application of minimal-dose RT approaches.

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Key Points

Age-related declines in muscle strength and power negatively influence healthspan independently of changes in muscle mass, highlighting the importance of preserving muscle strength and power across the lifespan.

Structured, progressive resistance training (RT) is effective for increasing muscle mass and improving strength, but participation in community-based RT interventions is low, highlighting the need to identify strategies that can improve engagement and adherence to such interventions.

Compared to traditional RT approaches, minimal-dose RT involving lower session volumes together with either (1) higher training intensities/loads performed at lower frequencies (low-volume, high-load RT) or (2) lower training intensities/loads performed at higher frequencies and with minimal-to-no equipment (resistance ‘exercise snacking’) may be more feasible approaches to improving muscle strength and function across the lifespan.

1 Introduction

Resistance training (RT) is the only non-pharmacological intervention known to improve skeletal muscle mass, strength, and power across all stages of the lifespan. Other forms of exercise, such as aerobic-type training (e.g. walking, running, cycling, swimming), which are often the focus of public health guidelines and campaigns [1], are less effective for increasing skeletal muscle mass [2] and strength [3] compared to RT. Even in those already meeting aerobic [i.e. moderate-to-vigorous physical activity (MVPA)] exercise guidelines, meeting muscle strengthening activity (including RT) guidelines independently reduces the risk of age-related losses of muscle mass, strength, and function [4]. While these observations highlight the independent benefits of muscle strengthening exercise on muscle mass and function, RT is also associated with many other health benefits [5] that are likely underappreciated compared to the health benefits of aerobic-based exercise. For example, RT is associated with reduced all-cause [6, 7] and cancer-related [7] mortality, and reduced incidence of various comorbidities including cardiovascular disease [8], musculoskeletal pain [9], anxiety and depressive symptoms [10], and high blood pressure [11]. Compared to aerobic exercise modalities, RT (particularly incorporating moderate-to-high loads) also has beneficial effects on bone health, which reduces the risk of

osteoporosis and fractures and associated morbidity and mortality [12]. RT can also improve both cardiorespiratory fitness (in previously untrained individuals) [13], skeletal muscle mitochondrial content and function [14, 15], and glycaemic control in people with type 2 diabetes [16], all of which are responses typically associated with aerobic-based exercise. For these reasons, RT has been suggested as a cornerstone strategy for improving healthspan [5] and should arguably be prioritised over other forms of exercise (e.g. aerobic exercise) where required, particularly in adults aged 45–50 and older in whom muscle mass, strength, and functional ability begin to decline [17].

Given the overwhelming evidence supporting the health-related benefits of RT, current global physical activity guidelines recommend adults (aged ≥ 18 years) perform muscle strengthening activities (i.e. use of weight machines/body weight exercises) at least twice per week [1]. Despite the established benefits of RT, only 10–30% of adults report meeting current muscle strengthening activity guidelines [18], a figure reported as low as 6% in Australian adults aged 50 and older [19]. Such poor engagement with RT is likely due to numerous potential barriers including time constraints, a high perceived difficulty and/or complexity, and limited access to facilities and equipment [20, 21]. Identification of RT strategies that limit barriers to participation and are effective for maintaining or improving muscle health and function will likely improve participation rates, and in turn, improve population health outcomes. As a result, there has been interest in identifying effective, pragmatic, and time-efficient RT strategies, which may be performed at home or in other remote environments, for preserving strength and functional ability across the lifespan. Emerging evidence suggests minimal-dose RT, characterised by lower session volumes performed at similar or higher weekly frequencies (with lower and higher frequencies associated with RT performed with higher and lower intensities/loads, respectively) than recommended in clinical consensus and public health guidelines, can improve strength, muscle mass, and functional ability in both younger and older adults [22–32]. The potential health-related benefits of minimal-dose exercise are highlighted by findings that MVPA bouts as short as 5 min are associated with lower mortality [33], while as little as 15 min of daily MVPA reduces mortality risk (by 14%) compared with no MVPA [34]. Minimal-dose exercise approaches are also consistent with recent shifts in physical activity guidelines suggesting any physical activity is better than none [1], and the greatest benefits of physical activity are realised when individuals progress from being inactive to performing some level of activity (i.e. physically active, but not meeting current guidelines) [35]. This narrative review therefore aims to (1) summarise the evidence for minimal-dose RT as a strategy for preserving muscle mass, strength, and functional ability with ageing; (2) propose

minimal-dose RT as a potentially more feasible strategy for improving healthspan compared to traditional RT; and (3) discuss practical models and associated considerations for the application of minimal-dose RT approaches.

2 Muscle Strength and Power Influence Healthspan Independently of Muscle Mass

Changes in aspects of skeletal muscle function, including strength and power, occur across the lifespan due to various physiological factors [36] and significantly impact overall health and quality-of-life [37, 38]. Indeed, ageing is associated with progressive losses of skeletal muscle mass, strength, power, and functional ability [39]. Sarcopenia, a condition characterised by age-related declines in muscle mass, strength and function, is estimated to affect ~30% of community-dwelling older adults aged over 65 [40] and ~50% of those aged over 80 [41]. This is of high clinical importance, as sarcopenia has been shown to predict reduced quality-of-life and difficulty performing daily activities and is associated with an increased risk of falls, fractures, institutionalisation, and mortality [37, 38]. The economic burden of sarcopenia is substantial, with direct health care costs estimated at US\$18.5 billion in 2000 [42], and causes a tripling of individual health care costs compared to those without sarcopenia [43]. Age-related losses of muscle mass, strength, and function are accelerated during short-term periods of imposed physical inactivity [44], such as during illness or hospitalisation, where older adults may spend more than 80% of time in bed, perform less than 750 steps daily [45] and can experience a loss in leg lean mass of approximately 1 kg after just 3–10 days [46]. Such short-term periods of reduced activity are therefore particularly critical to target for intervention against the detrimental effects of muscle loss and sarcopenia.

While older adults aged ≥ 60 years are estimated to lose between 3.7 and 4.7% (median values for females and males, respectively) of their peak muscle mass per decade [39], indices of strength and power decline between two to eight times faster [47, 48], and losses of muscle mass only explain a small fraction (6% and 8% in males and females, respectively) of the between-subject variance in loss of knee extensor strength [49]. In addition, while both low muscle mass (estimated using the skeletal muscle index (SMI) [50]) and muscle function (assessed by 30-s sit-to-stand performance) increased the risk of losing physical independence in community-dwelling older adults (mean age 75 years), the risk was almost four times greater in those with low muscle function [odds ratio 6.19, 95% confidence interval (CI) 5.08–7.53] than individuals with low muscle mass (odds ratio 1.65, 95% CI 1.27–2.31) [51]. Together, these observations indicate a clear dissociation between age-related loss

of muscle mass and declines in muscle strength, power, and functional ability. This is particularly concerning since low muscle strength and power, but not low muscle mass, are more strongly associated with falls, fractures, and mortality [52]. Along with other age-related physiological (e.g. sensorimotor) changes, declines in muscle strength and power largely underpin impairments in functional task performance (e.g. gait speed, stair ascending/descending, rising from a chair) with ageing [53, 54], which substantially influences independence and quality-of-life. These data suggest that exercise-based strategies focused on increasing muscle mass, which often require greater RT volumes coupled with higher intensities of effort [55], may not be necessary for improving functional task performance and the capacity to perform activities of daily living. Rather, strategies to maximise healthspan should arguably focus on preserving muscle strength, power, and function rather than on attenuating losses of muscle mass per se. Notwithstanding, the many physiological and metabolic benefits of skeletal muscle mass, which serves as an important reservoir for glucose disposal and source of amino acids in times of need (e.g. periods of fasting), must be acknowledged [56]. While it is therefore likely important to attenuate age-related losses of skeletal muscle mass to preserve these metabolic benefits, we contend that a focus on specifically addressing age-related losses of muscle strength and power, which may occur somewhat independently of changes in muscle mass, with RT will have greater implications for overall health and functional ability with ageing.

Specifically targeting age-related losses of muscle strength and power with appropriate exercise prescription firstly requires an understanding of the physiological stimuli for improvements in these parameters. To ensure these strategies are effective in practice, an understanding of how these stimuli may be achieved in both a feasible and more enjoyable manner, which together may improve adherence to such interventions, is required to improve population health outcomes. The following section will firstly summarise evidence-based recommendations for improving muscle strength, power, and muscle mass with traditional RT. The evidence for the efficacy of minimal-dose RT approaches for improving muscle mass, strength, and power will then be summarised, before proposing minimal-dose RT as a strategy for preserving these capacities (particularly muscle strength) across the lifespan.

3 Minimal-Dose Resistance Training for Improving Muscle Strength and Power

Traditional, structured, and progressive RT is a well-accepted strategy for improving muscle strength, power, and muscle hypertrophy. The available evidence suggests that

with structured progressive RT, strength is best developed with nearer-to-maximum loads [e.g. $\geq 85\%$ one repetition maximum (1RM)] using exercises that closely mimic the test in which strength is measured, with inter-set rest intervals of > 2 min and a training volume of < 15 sets per muscle group per week [55]. Optimal RT frequencies for developing strength are not known, although RT frequency per se is not considered an independent determinant of strength gain [55]. In contrast to key stimuli for strength development, ‘intensity of effort’, defined as the number of repetitions performed as a function of the maximum possible with a given absolute load, or the proximity to the point of volitional or momentary muscular failure, has been proposed as an important stimulus for skeletal muscle hypertrophy [55]. It is therefore well-accepted that muscle hypertrophy is possible when RT is performed with a wide variety of loads (i.e. 30% to $> 70\%$ 1RM), providing intensity of effort is high [55] (but not necessarily maximal [57]). Other determinants of muscle hypertrophy appear to be training volume (particularly for trained individuals), inter-set rest periods of at least 30 s, and a training frequency of at least two sessions per week [55].

Despite the well-accepted efficacy of RT for eliciting multiple health benefits, engagement and adherence to traditional RT strategies is typically poor (~ 10 – 30% of the population meeting current guidelines) [18, 19]. This suggests alternative approaches are needed, particularly those that prioritise muscle strength and power development. Like any targeted exercise prescription, strategies to promote specific physiological adaptations may be informed by an understanding of the physiological stimuli of these adaptations, and how to elicit these stimuli via the appropriate manipulation of RT variables. The physiological stimuli for muscle hypertrophy and strength development are indeed somewhat distinct, which may facilitate selective development of these adaptations with appropriate exercise prescription. For example, the application of mechanical load (tension) appears important for both strength development and muscle hypertrophy, whereas higher training volumes performed with sufficient intensities of effort are more important stimuli for muscle hypertrophy [55]. In this manner, RT prescription may be tailored to selectively develop muscle function with or without considerable improvements in muscle mass (i.e. to prioritise development of strength and power relative to body mass).

As well as influencing RT prescription strategies, the somewhat distinct stimuli for strength and muscle hypertrophy development may also have important implications for adherence to these strategies at the population level. For example, it is possible the additional stimuli needed to develop muscle hypertrophy as well as strength—namely the combination of higher intensities of effort and overall RT volumes—may promote additional barriers to RT participation. Indeed, higher RT volumes and higher intensities

of effort are associated with greater perceptions of discomfort [58] and less enjoyment [59] and require a greater time commitment [58], all of which may negatively influence long-term adherence to RT [58–60]. While the contribution of muscle hypertrophy per se to strength development with RT has been the subject of debate [61, 62], it is clear that substantial strength improvements can be achieved with minimal doses of resistance exercise, which can occur in the absence of considerable muscle hypertrophy [28, 29]. Given the importance of preserving muscle strength and power across the lifespan, and some of the challenges associated with undertaking RT with higher volumes and intensities of effort, pragmatic and feasible RT strategies for maximising healthspan should arguably emphasise minimal-dose approaches focused on improving muscle strength and power. The following section will provide an overview of various models of minimal-dose RT and discuss some associated practical considerations for their implementation.

4 Characterising Minimal-Dose Approaches to Resistance Exercise

While this review focuses on resistance-based exercise, to date, the majority of research on minimal-dose exercise approaches has concentrated on aerobic-based exercise modalities. Nevertheless, some of the findings gleaned from these studies may inform the characterisation of minimal-dose resistance-based exercise. In their review of the physiological basis of minimal-dose approaches of aerobic-based exercise modalities (e.g. cycling or stair climbing) for improving health, Gibala and Little [63] distinguish between ‘brief, vigorous exercise’ and ‘exercise snacks’. Briefly, vigorous exercise was defined as either a continuous bout of vigorous-intensity exercise (defined according to American College of Sports Medicine (ACSM) guidelines [64]) lasting ≤ 5 min, or in the case of intermittent exercise, such as high-intensity interval training (HIIT), no more than 5 min of cumulative vigorous exercise (e.g. 5×1 min intervals). Alternatively, ‘exercise snacks’ were considered as multiple shorter bouts (~ 20 s) of vigorous exercise that could be incorporated into the day with several hours of rest in between. Therefore, the main distinguishing features of ‘exercise snacks’ versus ‘brief, vigorous exercise’ appear to be shorter exercise bouts performed with minimal-to-no equipment, which may be undertaken repeatedly (or more than once) throughout a given day and/or more frequently during a given week. In the context of aerobic exercise, both ‘brief, vigorous exercise’ and ‘exercise snacks’ may also be scaled based on equipment availability to incorporate either traditional (e.g. cycling ergometer) or non-traditional/minimal equipment (e.g. stair climbing) modalities.

The definitions of ‘brief, vigorous exercise’ and ‘exercise snacks’ applied to aerobic-based exercise [63] may be adapted for resistance-based exercise. For example, brief, vigorous exercise may be considered analogous to structured, low-volume (e.g. no more than one to two working sets per muscle group per session) and high-load (e.g. $\geq 70\%$ 1RM) RT performed with application of external loads and/or equipment (such as barbells/dumbbells or machines). Alternatively, resistance ‘exercise snacks’ may be considered as resistance exercise performed using either bodyweight or minimal equipment (e.g. resistance bands or portable weights) and involving briefer, but more frequent, exercise bouts compared with a low-volume, high-load RT approach (Fig. 1). Together these factors make a resistance ‘exercise snacking’ (or ‘strength snacking’) approach more suitable for being performed remotely (in non-gym settings) and repeatedly throughout a given day and likely using a time-based (rather than a relative intensity- or load-based) prescription, which may also improve ease of prescription and application of this approach when performed remotely and/or unsupervised.

Similar to the analogous aerobic exercise approaches, minimal-dose RT may also involve either traditional (e.g. barbells/dumbbells or machines) or non-traditional/minimal equipment (e.g. bodyweight, resistance bands, portable weights) modalities. Due to differences in equipment requirements and inherent suitability for being performed repeatedly throughout a given day, minimal-dose RT strategies, considered as either ‘low-volume, high-load’ or resistance ‘exercise snacking’ approaches, may be more applicable to facilitate long-term adherence in gym-based and remote/home-based environments, respectively. However, an important question that will be addressed in the following section is whether ‘low-volume, high-load’ and resistance ‘exercise snacking’ minimal-dose RT approaches are effective for improving muscle strength and muscle hypertrophy.

5 Efficacy of Minimal-Dose Resistance Training Approaches for Improving Muscle Mass, Strength, and Power

Compared to aerobic-based exercise, there is less evidence for the efficacy of minimal-dose resistance-based exercise approaches for improving muscle strength, power, or muscle hypertrophy. Several studies have determined the influence of ‘exercise snacking’ approaches focused on resistance/strength-based exercise on short-term changes in markers of cardiometabolic health, particularly in the context of limiting sedentary or sitting time [65, 66]. Nevertheless, there is emerging evidence for the efficacy of minimal-dose RT, characterised as either low-volume, high-load RT

or resistance ‘exercise snacking’, for improving indices of strength, power, or muscle hypertrophy.

5.1 Low-Volume, High-Load Resistance Training

Several studies have determined the efficacy of low-volume RT performed with either high ($\geq 85\%$ 1RM) or maximal (i.e. at or very close to the 1RM) loads on improvements in strength and muscle hypertrophy, with or without comparison to higher volumes of lower-load RT [22–29, 32, 67, 68].

Using a unilateral exercise design, Dankel et al. [28] had younger males (mean age 25 years) with a minimum of 1 year of RT experience perform both a unilateral 1RM and isometric maximal voluntary contraction (MVC) test for the elbow flexors on both arms once per day for 21 consecutive days. In addition to the strength testing, one arm also performed additional low-load resistance exercise (3×10 repetitions at 70% 1RM) each day [28]. One condition therefore involved a low-volume dose of high-load RT each day, while the other received the same RT dose combined with a higher dose of lower-load RT. Improvements in 1RM strength were similar for the lower- ($+1.9$ kg) and higher-volume ($+2.2$ kg) groups, although isometric strength was not improved in either condition, while muscle hypertrophy (muscle thickness assessed via ultrasound) was evident only in the higher-volume condition. A follow-up study [68] in which RT was performed more frequently (i.e. twice per week) and consistent with current muscle strengthening exercise guidelines [1] also found similar improvements in both isometric (mean difference 0.18 N) and dynamic 1RM strength (mean difference 0.28 kg) of the elbow flexors for low-volume, high-load RT (a single maximal-intensity repetition achieved within five attempts) compared with traditional higher-volume, lower-load RT (four maximal-repetition sets with an 8 – 12 RM load).

Taken together, these findings [28, 68] suggested the improvements in strength were related to specificity between the training intervention method of strength assessment used (a notion termed ‘practising the test’) and were not augmented by additional RT volumes or muscle hypertrophy. The observations therefore demonstrated an important proof-of-concept that even minimal doses of high-load resistance exercise over a short period can improve dynamic (i.e. 1RM) muscle strength in resistance-trained individuals.

To determine improvements in strength and muscle hypertrophy with low-volume, higher-load RT compared with higher-volume, lower-load RT involving both the upper and lower body, Mattocks et al. [29] had younger (mean age 21 – 22 years) untrained males and females perform eight weeks of RT (two sessions per week) in which participants undertook either low-volume, high-load RT (5×1 RM attempts separated by 90 s of recovery) designed to maximise strength independent of muscle hypertrophy,

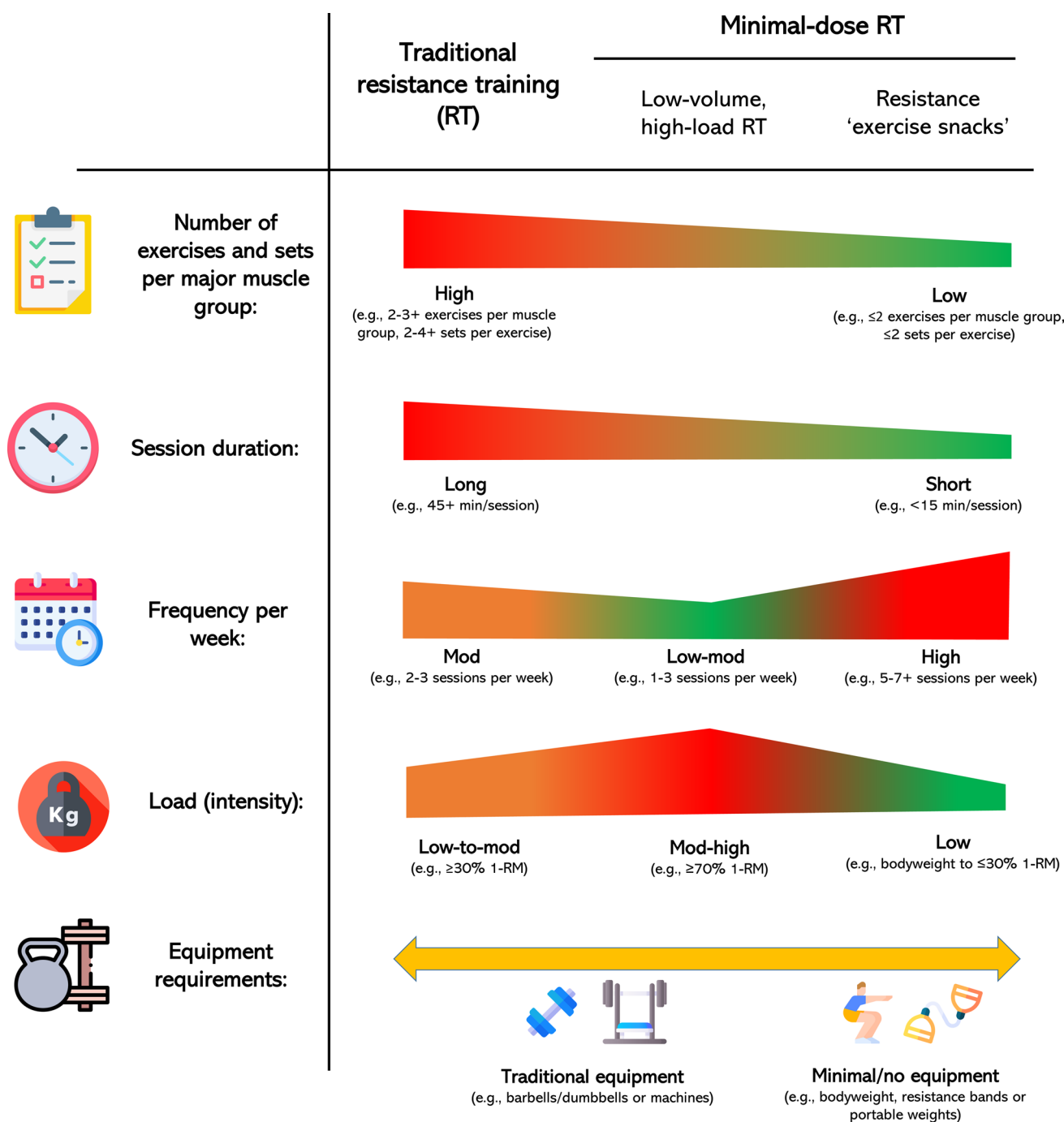


Fig. 1 Characteristics of minimal-dose resistance training (RT) approaches (either low-volume, high-load RT or resistance 'exercise snacks') as compared to traditional RT. Compared to a typical programme of traditional RT involving relatively high volumes per session (e.g. 2–3+ exercises per major muscle group and 2–4+ sets per exercise) performed at a moderate exercise frequency (e.g. 2–3 sessions per week), minimal-dose RT approaches are characterised

by lower volumes per session (e.g. ≤2 exercises per major muscle group and ≤2 sets per exercise) in combination with either (1) lower weekly training frequencies and higher intensities/loads (for low-volume, high-load RT) or (2) higher weekly training frequencies and lower intensities/loads (for resistance 'exercise snacking' approaches) requiring minimal-to-no equipment (i.e. bodyweight, resistance bands, or portable weights). *1-RM* one repetition maximum

or higher-dose, lower-load training (4 × 8–12 repetitions separated by 90 s of recovery) designed to simultaneously increase strength and muscle mass. Similar to previous

findings [28], dynamic 1RM knee extension (2.7 vs. 3.7 kg, respectively) and chest press (4.4 vs. 2.5 kg, respectively) strength gains (and isometric or isokinetic knee extension)

were equivalent for the lower-volume, higher-load and higher-volume, lower-load conditions, despite greater muscle thickness gains of the posterior upper arm (0.25–31 mm) and anterior upper leg (0.25–0.36 mm) in the higher-volume group. These findings add further support to the notion that minimal doses of high-load resistance exercise (involving both the upper and lower body) can improve strength, and higher doses of resistance exercise likely confer greater benefits for muscle hypertrophy, but not strength per se.

While low-volume, high-load RT can improve strength in both untrained individuals and those with RT experience [28, 29, 68], it is possible these approaches may become less effective as strength levels improve over time. Nevertheless, minimal-dose RT (specifically low-volume, high-load RT) is also effective for improving strength in highly strength-trained individuals. For example, strength improvements of ~6–11% were achieved in well-trained powerlifters and weightlifters with low volumes of very-high load RT (1RM squat followed by 5 × 3 repetitions at 85%, or 5 × 2 repetitions at 90%, of the daily 1RM) performed consecutively for 37 days [67]. In another case study [32], beginner-intermediate powerlifters performed 10 weeks (three sessions per week) of either low-volume, very high-load RT (1 × 1 repetition very close to the 1RM) or a traditional, higher-volume and lower-load periodised approach (3–15 × 1–7 repetitions at intensities between 55 and 93% 1RM). The minimal-dose approach equated to ~10% of the total training dose of the higher-volume approach. Two out of the three participants that performed the traditional, higher-volume approach improved their muscle strength (indicated by an improved combined total load lifted during the squat, deadlift, and bench press exercises), while two of the five participants who performed the low-volume, very high-load training improved their muscle strength and three experienced a decrease in their strength. While these findings suggest low-volume, very high-load RT can improve muscle strength in well-trained individuals, this approach may lead to performance decrements in trained individuals when performed for extended periods, and is perhaps most effective during shorter training phases since most strength improvements occurred in the initial five weeks of training [67]. Nevertheless, it is important to highlight that low-volume, high-load RT can improve strength even in most well-trained strength and power athletes.

While low-volume, high-load RT is clearly effective for improving muscle strength, the high intensities ($\geq 85\%$ 1RM) used may be impractical in untrained individuals or in remote exercise interventions where access to equipment is limited. Comparatively higher volumes of lower, albeit still relatively high-load (e.g. ~70–85% 1RM), RT may therefore be a more feasible alternative in those with limited RT experience, and/or as a periodic 'de-load' from periods of very high-load training. Few studies have investigated whether

low volumes (achieved via single sets per exercise) of moderate-to-high-load RT can improve indices of muscle mass, strength, and function [22–27]. For example, Westcott et al. [24] evaluated whether 10 weeks of combined aerobic and resistance exercise, at doses consistent with ACSM recommendations (20 min of aerobic exercise three times per week and one set of eight to 12 repetitions of eight to ten resistance exercises twice per week) [69] performed at varying frequencies (one, two, or three times per week), influenced changes in body composition and blood pressure in adults aged 21–80 years (mean age 55 years). The exercise protocol involved ~20 min of aerobic exercise (steady-state running/walking or cycling) and ~20 min of resistance exercise (consisting of one set of ten exercises performed for between eight and 12 repetitions to volitional fatigue). While larger training doses (frequencies) produced greater fat mass loss (i.e. 3 days/week was better than either 1 or 2 days/week), changes in lean mass were no different between the 2 and 3 days/week groups (but greater than 1 day/week). These findings suggest single sets of moderate-to-high-load RT performed with low frequencies (minimum 2 days/week) can improve lean body mass, although indices of muscle strength and power were not assessed in that study [24].

In another study employing only low-volume, high-load RT (without additional aerobic exercise), Fisher and colleagues [23] had middle-aged older adults (aged ~55 years) perform brief (~12 min per session) and infrequent (twice weekly) RT involving a single set of five to seven multi-joint exercises. Each set was performed with the self-determined 6RM load (i.e. the load that predicted muscular failure on the 7th repetition). After ~12–19 weeks of training, mean muscle strength improvements (assessed by the progression in load lifted per exercise) ranged from 38 to 91%. Another investigation by Steele et al. [22] showed that 6 months of RT, involving a single set of no more than 12 repetitions (i.e. the self-determined 12RM) for seven exercises, improved strength (up to 24%), body composition assessed via bio-electrical impedance analysis (BIA) (i.e. fat mass, fat percentage, and muscle percentage, but not muscle mass), and multiple functional task measures (e.g. stair climbing, a carrying task, and chair rise) in older males and females (mean age 69 ± 5 years). It is again important to highlight that these improvements in muscle strength and functional task performance occurred without substantial changes in muscle mass.

As well as improving measures of muscle strength, minimal doses of high-load RT also positively influence bone parameters. For example, 8 months of twice-weekly, high-load RT (5 × 5 repetitions at 80–85% 1RM for the squat, deadlift, and overhead press) combined with a single weight-bearing impact loading activity (jumping chin-ups with drop landings) in postmenopausal women (mean age 65 years) with osteopenia/osteoporosis improved lumbar spine and femoral neck bone mineral density (BMD)

and femoral neck cortical thickness relative to controls [25]. The positive effects on bone parameters occurred alongside improvements in muscle strength (knee and back extensors), power (vertical jump), and functional task performance (timed-up-and-go, five times sit-to-stand, and functional reach). A subsequent study [70] found an identical low-volume and high-load RT protocol increased femoral neck medial cortical thickness and various tibial bone parameters versus controls, and positively influenced various other bone parameters, including femoral neck and total hip trabecular and total volume. While the aforementioned high-load RT interventions can positively influence bone parameters and aspects of neuromuscular function [25, 70], it is possible this form of RT is not suitable for all individuals. For example, 101 of the 406 women (~25%) assessed for eligibility in one study [25] were excluded due to contraindications to performing the RT protocol, suggesting this form of higher-load RT may not always be feasible for specific population groups.

In addition to musculoskeletal benefits, there is also evidence that low-volume, moderate-to-high load RT can improve cardiometabolic health markers. In young (mean age 36 years) healthy (insulin-sensitive) but overweight males, 6 weeks of RT involving a single set of nine exercises performed to failure (80% 1RM), taking 15–20 min to complete, improved insulin sensitivity by 16%, as well as measures of strength (16–28%), power (~20–31% increase in knee extensor rate of torque development), and muscle size (~10% increase in vastus lateralis muscle thickness) [26]. Collectively, the current evidence [22–27] suggests minimal doses of structured, moderate-to-high-load RT can improve measures of muscle mass, strength, and power, metabolic health, and bone parameters, in adults across a wide age spectrum.

While low-volume, moderate-to-high-load RT is effective for improving a wide range of physiological parameters linked to healthspan [22–27], adherence to such interventions is critical to their real-world effectiveness. An important observation from Steele et al. [22] was that after the 6-month period of supervised, moderate-to-high-load RT, only 57% of participants elected to continue performing unsupervised RT for a further 6 months, despite the same access to facilities and equipment. This reduction in compliance reversed the improvements in muscle mass to baseline values and reduced strength values (up to ~24% depending on the exercise) at the 6-month follow-up, although strength remained above baseline values at this time point [22]. Therefore despite the efficacy of this form of supervised, structured minimal-dose RT, the continuation of unsupervised RT offered limited benefit versus training cessation alone. Further work is needed to determine whether there are potential benefits of alternative RT

strategies that offer greater flexibility in terms of equipment, location, and time-of-day scheduling.

5.2 Resistance ‘Exercise Snacking’

The concept of ‘exercise snacking’ has been explored as a time-efficient option to improve health outcomes [65, 66, 71, 72], but there is less evidence for the efficacy of ‘exercise snacking’ resistance exercise approaches for improving muscle mass, strength, power, or functional task performance. Most ‘exercise snacking’ approaches applied to resistance-based exercise have focused on their efficacy for short-term changes in markers of cardiometabolic health, such as lowering postprandial glycaemia [71, 72] and for interrupting periods of prolonged sitting [65, 66]. To our knowledge, only one study has investigated the influence of ‘exercise snacking’ based solely on resistance exercise on changes in muscle mass, strength, power, and/or functional task performance [30]. This pilot study [30] determined the efficacy of a 28-day ‘exercise snacking’ protocol for improving strength, body composition, and functional ability in inactive older community-dwelling males and females (mean age 70 years). The ‘exercise snacking’ protocol involved five exercises, each performed as many times as possible for 1 min, with 1 min of recovery between exercises, undertaken twice per day for 28 consecutive days. This exercise protocol improved 60-s sit-to-stand performance (+31% for the intervention group vs. no change for control), but not leg press muscle force (+5% and –2% for intervention and control, respectively) and power (+6% and –2% for intervention and control, respectively), although moderate-to-large effect sizes (Hedges’ $g = 0.49$ – 0.81) for the between-group differences (versus control) were noted. The ‘exercise snacking’ intervention did not change total or lower-body lean mass assessed via dual energy X-ray absorptiometry (DXA), although there was a moderate effect size (Hedges’ $g = 0.68$) for the difference in the change in lower-body lean mass versus controls. Similarly, thigh muscle cross-sectional area [assessed via peripheral quantitative computed tomography (pQCT)] was unchanged within both groups, although there was a large effect size (Hedges’ $g = 0.96$) for the between-group difference in the change score. These findings, coupled with excellent adherence (98%) and lack of adverse events, indicated that ‘exercise snacking’ may be a promising approach for improving indices of muscle strength and function in older adults.

Although not exclusively focused on resistance exercise, another study [31] determined the effects of 4 weeks of remotely delivered, brief (6 min per session), and frequent (either once- or twice-daily) high-intensity bodyweight circuit training on body composition, strength, cardiorespiratory fitness, and quality-of-life in young (mean age 25 years) untrained males and females. The results showed

that once- and twice-daily protocols were similarly effective for improving the maximum number of repetitions possible in various bodyweight exercises (push-ups, leg levers, burpees, 45° one-legged squats, and 30-s skipping), lowering heart rate during sub-maximal exercise, and improving quality-of-life [31]. Neither the once- nor twice-daily protocols altered sub-maximal or peak oxygen uptake, the respiratory exchange ratio, or body composition. There was excellent adherence to both the once- and twice-daily protocols (90.7% and 85.7%, respectively). These findings lend further support to the notion that brief, simple, and frequent exercise interventions consistent with an ‘exercise snacking’ approach are feasible and can improve aspects of muscle function independently of changes in body composition.

A key practical consideration of ‘exercise snacking’ strategies with strong implications for both the feasibility and scalability of this approach is the requirement of minimal (e.g. resistance bands, small portable weights, or household items) to no equipment. Importantly, there is evidence for comparable improvements in strength and functional task performance when RT is performed using bodyweight only [73, 74] or minimal equipment (e.g. resistance bands [75]) compared with traditional RT equipment. These findings also extend to exercises whereby the influence of bodyweight loading is minimal. For example, in an interesting proof-of-concept study, comparable improvements in muscle mass (albeit lower strength gains) were evident with RT of the elbow flexors incorporating no external loading (only forceful muscle contraction throughout the range of motion) compared to high-load (70% 1RM) RT [76]. Since access to equipment and facilities presents a significant barrier to engagement with RT [20], these observations highlight the potential benefits of ‘exercise snacking’ approaches for engagement in resistance exercise compared with traditional approaches.

It is also worth noting the shortened recovery duration between repeated exercise sessions (owing to a higher training frequency) typically associated with ‘exercise snacking’ approaches may not negatively influence changes in strength and power compared with longer between-session recoveries typically recommended for traditional RT approaches (typically ≥ 48 h between sessions involving the same major muscle groups). For example, RT performed on consecutive days (three sessions performed on three consecutive days per week) is comparably effective for improving muscle strength as performing the same number of sessions interspersed by 48–72 h recovery over 12 weeks [77]. These weekly training frequencies (three sessions per week) are, however, lower than may be performed using an ‘exercise snacking’ approach (which may be performed multiple times daily), so further evidence is required to determine optimal training frequencies and/or between-session recovery durations when employing an ‘exercise snacking’ approach.

Taken together, there is emerging evidence to support the notion that brief and frequent ‘exercise snacking’ approaches to resistance exercise are feasible and efficacious for improving strength and power (and potentially muscle mass). These benefits occur despite ‘exercise snacking’ approaches requiring minimal-to-no equipment and limited external loading, which may otherwise pose barriers to RT participation.

6 Physiological Bases for the Potential Benefits of Minimal-Dose Resistance Training

While there is evidence that only minimal doses of RT are required to improve strength, which may occur independently of changes in muscle mass, the physiological bases for these benefits are emerging and will be discussed briefly below.

6.1 Physiological Bases for the Benefits of ‘Low-Volume, High-Load’ Resistance Training

Neural adaptations, including enhanced muscle activation, faster motor unit recruitment and firing rates, and improved coordination, are important mediators of improvements in strength and power with RT [78]. On the contrary, the role of muscle hypertrophy per se in strength and power development with RT is less clear [61, 62]. Since neural adaptations that mediate improvements in muscle function are maximised when RT is performed with higher versus lower loads [79], it is possible that low-volume, higher-load RT approaches, particularly when performed with faster intended movement velocities [80], facilitate neural adaptations that improve strength and power largely independent of the potential contribution of muscle hypertrophy (Fig. 2).

Because neural adaptations (and by extension, strength gains) are highly task-specific [81], the efficacy of low-volume, high-load RT for improving strength also likely depends on the measures used to assess strength. In this manner, strength gains with low-volume, high-load RT (and indeed any RT intervention) are likely to be greatest when the measure(s) used to assess strength closely replicate the RT intervention itself—a notion termed ‘practising the test’ [28, 29, 82]. This has implications when interpreting studies comparing the efficacy of higher- versus lower-load RT for improving strength. For example, strength gains may be expected to be greater following low-volume, moderate-to-high-load RT compared with higher-volume, lower-load RT when strength is assessed using dynamic 1RM testing compared to isometric strength testing, since the former most closely replicates both the movement pattern and loading demands of high-load RT [28, 29, 82].

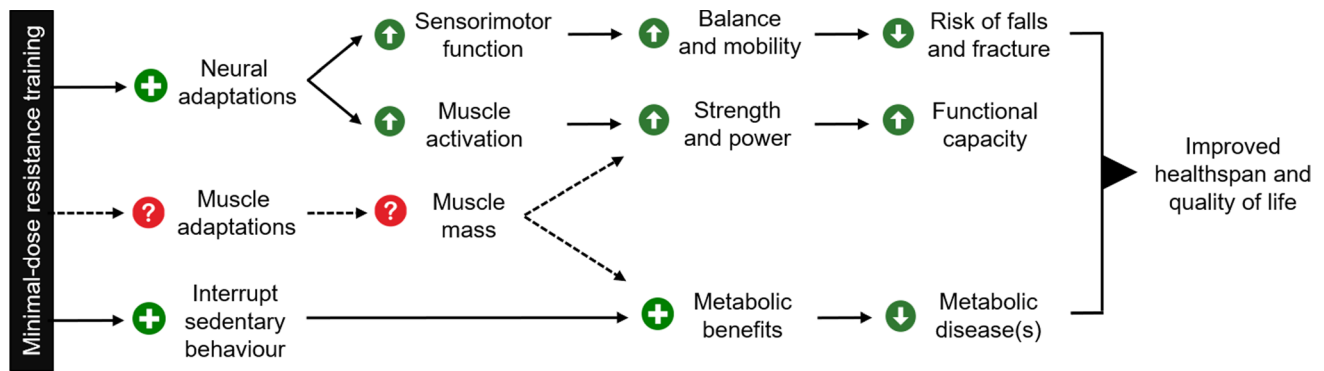


Fig. 2 Illustrative summary of the physiological (neural and muscle adaptations) and behavioural (interrupted sedentary behaviour) responses to minimal-dose resistance training and their relevance to

improvements in parameters that ultimately increase healthspan and improve quality-of-life with ageing

While conclusions on the efficacy of lower-volume, higher-load RT (and consequently any comparisons with other RT strategies) for improving strength may be influenced by the measure of strength used, the extent to which this negates the potential benefits of these approaches is unclear. For example, given strength is a desirable quality associated with greater independence and quality-of-life [83], and lower risk of both morbidity and mortality [52], it is unclear whether the method by which strength is assessed influences the health-related benefits of improved strength. This concept requires further examination, particularly in long-term intervention studies examining strength improvements with RT. Nevertheless, it may be important that future studies assess changes in strength using multiple measures, and include strength tests that are unfamiliar to all intervention groups [82]. In addition, while there is evidence that levels of strength may be more important than RT participation per se (i.e. meeting muscle strengthening activity guidelines) for reducing the risk of premature all-cause mortality, further work is required to determine whether performing RT with the goal of exploiting strength outcomes (e.g. by ‘practising the test’ and/or by using minimal-dose RT approaches such as low-volume, high-load RT) negates any of the benefits associated with participation in more traditional RT approaches.

In summary, strength gains with low-volume, high-load RT are likely attributed to neural adaptations that may improve strength independent of muscle hypertrophy, together with the typically high degree of similarity between high-load RT interventions and measures used to assess changes in strength.

6.2 Physiological Bases for the Benefits of Resistance ‘Exercise Snacking’

Compared to low-volume, high-load RT approaches, ‘exercise snacking’ approaches are typically characterised as

lower in intensity (pertaining to the absolute and relative loads lifted). Since the neural adaptations thought to mediate improvements in strength and function may be maximised with RT performed at higher absolute and/or relative intensities [79] and with high intended movement velocities [80], the potential benefits of resistance ‘exercise snacking’ for improving aspects of strength and power may be contingent on the higher training frequency of this approach and the subsequent influence of these higher frequencies on overall training volume. Current ACSM recommendations [84] suggest RT should be performed on 2 or more days per week, and that muscle groups should be trained at least 48 h apart. Compared to other means of applying progressive overload to stimulate physiological adaptations with RT, such as increasing the load being lifted, or the number of repetitions and/or sets (volume) performed in a given session, exercise frequency is an under-researched, and perhaps underappreciated, RT variable [85]. While progressive overload is undoubtedly important for stimulating physiological adaptation to exercise training, it is possible a theoretical threshold of exercise volume exists whereby additional volumes do not provide greater benefit for a given exercise session [85]. For this reason, additional RT volumes achieved by performing more repetitions or sets per muscle group in a given RT session may not yield greater benefits. In this instance, continued progressive overload (and therefore physiological adaptation) may be achieved by a higher training frequency associated with ‘exercise snacking’ approaches. It is therefore possible that the high frequencies associated with an ‘exercise snacking’ approach may facilitate adaptation by providing progressive overload attributable to higher training frequencies, even if individual sessions provide a suboptimal ‘dose’ of exercise for stimulating physiological adaptations. It should also be mentioned that ‘exercise snacking’ approaches may still allow for a high degree of intensity of effort to be performed, which is considered an important determinant of muscle hypertrophy

[55], and allow for the accumulation of reasonable training volumes if frequency remains high. For this reason, ‘exercise snacking’ approaches may, in addition to improvements in strength and function, also promote muscle hypertrophy, although current evidence is lacking in this regard.

The shorter, but more frequent, nature of ‘exercise snacking’ approaches likely have important benefits for interrupting periods of sedentary behaviour (Fig. 2). While less frequent but more structured and/or longer bouts of exercise are undoubtedly effective for improving markers of health and fitness, it is possible that individuals can meet physical activity guidelines despite being largely sedentary, consequently increasing their cardiometabolic risk profile [86]. Accumulating more frequent ‘exercise snacks’ could therefore be favourable for interrupting periods of sedentary behaviour that may otherwise occur around less frequent, structured exercise bouts. Importantly, a meta-analysis found such accumulated patterns of physical activity had comparable benefits for improvements in cardiorespiratory fitness and metabolic health markers (blood pressure), and even superior reductions in body mass and low-density lipoprotein cholesterol, compared to continuous exercise patterns [87], further highlighting the potential benefits of brief yet frequent exercise.

7 Behavioural Bases for the Potential Benefits of Minimal-Dose Resistance Training

Despite the established benefits of RT, less than 10% of the population meet current muscle strengthening guidelines [19], suggesting more work is needed to improve engagement with RT. Compared to traditional RT approaches consistent with current exercise guidelines [88], minimal-dose approaches may improve engagement with RT by limiting various barriers to participation. For example, lower-volume, higher-load RT is perceived as more enjoyable than higher-volume, lower-load RT in untrained younger males and females [59]. Such protocols are also associated with lowered perceptions of exertion and discomfort compared with higher-volume RT performed with lower loads [58], which may also positively influence long-term adherence.

The limited time commitment (per session) associated with minimal-dose RT may also improve the feasibility of RT by allowing sessions to be performed flexibly around other daily activities. It is also possible an increased emphasis on training frequency, as opposed to training quality and/or volume per session—consistent with ‘exercise snacking’ approaches—may have benefits for exercise adherence. For example, performing more frequent exercise sessions early in an exercise intervention promotes adherence to exercise in subsequent periods, which is consistent with habit

formation [89]. Specifically, one study found that for each additional exercise session performed per week over an initial 4-week period, the likelihood of exercising for each of the next 8 weeks increased by threefold [89]. While speculative, higher exercise frequencies (with limited session durations) may also lower the perceived difficulty of individual exercise sessions and in turn improve exercise self-efficacy (defined as the individual's belief in their ability to successfully perform a specific behaviour), which is a common barrier to engagement with RT [20, 21]. Despite these potential benefits, further work is needed to determine whether minimal-dose RT strategies are associated with better long-term engagement and exercise adherence compared with traditional RT strategies. It should also be acknowledged that for some individuals, higher exercise frequencies may pose an additional barrier to exercise participation, despite the brief nature of each exercise session. In these instances, a lower-frequency approach to minimal-dose RT (e.g. low-volume, high-load RT) may be a more feasible strategy.

In summary, the characteristics of minimal-dose RT approaches may improve participation and adherence to RT via multiple means, including by facilitating habit formation (particular for ‘exercise snacking’ approaches), allowing flexibility for exercising at a suitable time of day, and allowing for modification of the exercises performed depending on equipment availability and desired training frequency. A visual summary of the potential advantages of minimal-dose RT compared to traditional approaches for engagement and adherence to RT is presented in Fig. 3.

8 Practical Considerations for Maximising the Benefits of Minimal-Dose Resistance Training

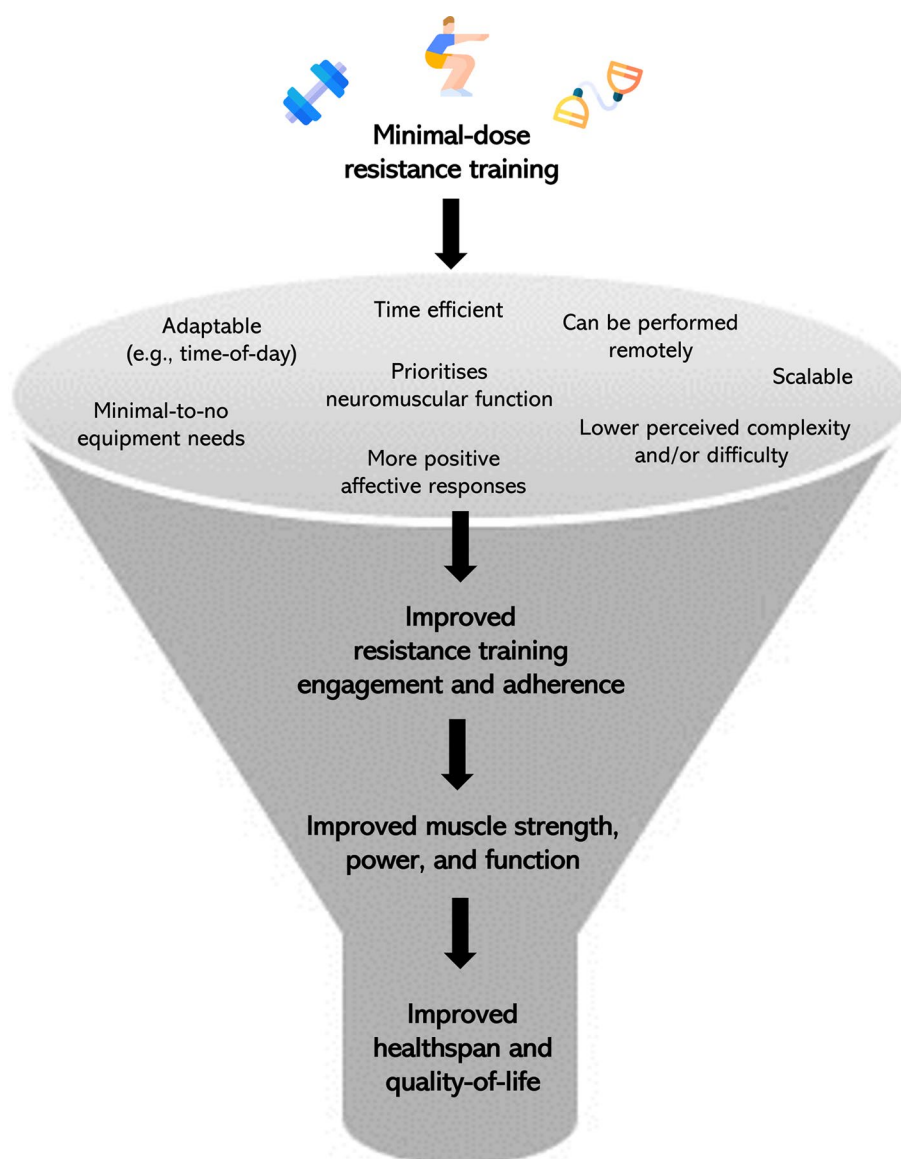
Both low-volume, high-load RT and resistance ‘exercise snacking’ provide a framework for the application of minimal-dose RT; however, there are additional considerations for maximising the potential benefits of these approaches. These considerations primarily involve exercise selection and the potential interactions of minimal-dose RT with other forms of exercise (e.g. aerobic exercise).

8.1 Exercise Selection

8.1.1 Prioritising Multi-joint Exercises

Given the brief nature of minimal-dose resistance exercise, the exercises incorporated into these approaches should be chosen with consideration of both time efficiency and specificity. Exercises involving movements associated with activities of daily living (e.g. squats, stair climbs, push-ups) that engage as many muscle groups as possible (thus

Fig. 3 Summary of the potential advantages of minimal-dose resistance training (RT) compared to traditional RT approaches for improving engagement and adherence to RT, leading to improvements in strength, power, and function that positively influence healthspan and quality-of-life



increasing time efficiency) should therefore be prioritised. For these reasons, multi-joint exercises should be favoured over single-joint exercises, as these exercises target more larger muscle groups in addition to the smaller muscles targeted in single-joint exercises. Lower-body exercises should also be favoured over upper-body exercises, as losses of muscle mass, strength, and function are more prominent in the lower versus upper body [90] and are therefore more likely to underpin declines in functional ability with ageing. Multi-joint exercises involving muscle groups and movement patterns more prominently

associated with activities of daily living may therefore involve squats, deadlifts, push-ups, lunges, or variations thereof.

8.1.2 Osteogenic Exercises

Along with muscle mass, bone mass also deteriorates with ageing, which is associated with an increased risk of fractures and associated mortality [12]. It is recognised that while progressive moderate-to-high load RT is more effective than other forms of exercise (e.g. low-impact or non-weight-bearing activities such as walking, swimming, or

cycling) for eliciting bone adaptations, certain exercises are more osteogenic (i.e. promote a positive skeletal response) than others. Specifically, exercises that are dynamic and associated with higher ground reaction forces applied rapidly and/or in novel directions are more osteogenic than static and slow movements [12]. Previous evidence suggests as little as 2–5 min of daily multidirectional hopping exercise (five sets of ten hops, with 15 s rest between each set) positively influenced various hip and proximal femur BMD parameters after 12 months of training [91, 92]. Where possible, minimal-dose exercise strategies should therefore incorporate osteogenic exercises, scaled to the level of individual risk of fracture [12], which target areas susceptible to fracture (hip, spine, and wrist) to improve bone outcomes. This may involve exercises that include jumping, hopping, drop jumps, plyometric push-ups, or scaled versions thereof.

8.1.3 Exercises that Challenge Balance

Declines in sensorimotor performance with ageing and subsequent deteriorations in balance and mobility are a substantial risk factor for falls [93] and the associated implications for fractures and mortality [94]. While traditional resistance exercises are effective for improving strength and/or muscle mass, they have mixed effects on functional performance measures such as balance and gait speed [95]. Minimal-dose resistance exercise should therefore incorporate some degree of challenging balance and dynamic mobility exercises, which, together with osteogenic exercises, may further reduce the risk of falls and fractures in older adults. Examples of such exercises may include single leg squats, rapid stepping, navigating an obstacle course, or movements performed on an unstable surface.

8.1.4 Exercises that Challenge Stabilising Muscles

The magnitude of declines in strength and power with ageing are muscle-group specific, with greater decreases in stabilising muscles (e.g. hip abductors and hip adductors) versus prime movers (e.g. knee extensors) [96, 97] and greater losses in the lower versus upper body musculature [90]. While prime movers are largely responsible for most locomotive tasks, stabilising muscles may contribute to both joint stiffening and in early responses to perturbation via feed-forward and/or feedback mechanisms, both of which influence stability and balance [98]. Accordingly, selection of exercises that engage the lower-body stabilising muscles, as well as those of the trunk [99], such as unilateral exercises (e.g. lunges, split squats) may further limit declines in the function of this musculature and improve outcomes related to functional task ability.

8.2 Combining Minimal-Dose Resistance and Aerobic Exercise Approaches

While others [5] have suggested RT should be prioritised over other forms of exercise (e.g. aerobic exercise), particularly for older adults, the combination of both resistance- and aerobic-based exercise likely promotes the greatest health benefits [100]. Combining resistance- and aerobic-based exercise (known as concurrent training), however, creates additional practical challenges, including not only a greater time commitment, but also the influence aerobic-based exercise can have on improvements in strength and power with RT—a concept known as the ‘interference effect’ [101]. Specifically, integrating both resistance and aerobic training into an exercise programme can lessen the development of strength and power relative to undertaking RT alone [102]. The ‘interference effect’ has been observed in some (e.g. [103, 104]) but not all (e.g. [105, 106]) studies, likely due to between-study methodological variations, including the exercise interventions used and other non-training-related factors such as participant training status and the measures used to assess changes in strength or power [107].

Based on our current understanding of the factors mediating the ‘interference effect’ with concurrent training, a number of practical recommendations for combining minimal-dose resistance and aerobic exercise approaches may be derived [107]. For example, since the interference effect may be at least partially mediated by the effects of residual fatigue from endurance training on RT performance [108], undertaking resistance and aerobic training sessions separately, either in morning/afternoon splits or on alternate days, may limit any potential ‘interference effect’. When performing both resistance and aerobic training within the same exercise session, which is a common approach to maximise time efficiency, RT should therefore precede aerobic training to minimise any potential ‘interference effect’ attributable to residual fatigue [108]. While further work is needed to determine the efficacy of minimal-dose approaches to both aerobic and RT in isolation, additional research is required to determine strategies to combine these two modalities in a manner that minimises any potential interference effect. This evidence may then directly inform practical strategies to maximise the benefits of both resistance and aerobic training in a manner that is both feasible and time efficient.

9 Conclusions and Future Research Directions

While skeletal muscle mass declines with ageing, losses of strength and power occur up to two and eight times faster [47, 48] and are stronger independent predictors of morbidity and all-cause mortality. For these reasons, strength

and power are closely linked to healthspan, and interventions that improve these parameters, with or without concomitant improvements in muscle mass, may substantially influence health and quality-of-life. While RT is the only known non-pharmacological strategy to improve muscle mass, strength, and power, engagement and adherence to traditional resistance exercise interventions is poor. There is growing evidence that minimal-dose approaches to RT can improve indices of muscle strength and power. Such minimal-dose RT approaches are characterised by relatively brief sessions of RT involving either (1) moderate-high loads performed infrequently (i.e. low-volume, high-load RT) or (2) lower loads performed at relatively higher frequencies and with minimal-to-no equipment (i.e. resistance ‘exercise snacking’ approaches). Minimal-dose approaches to resistance exercise therefore have the potential to limit or remove various barriers to RT participation, and likely have various positive implications for the potential scalability of, and adherence to, such interventions. In addition, resistance ‘exercise snacking’ approaches involving higher training frequencies may be beneficial for interrupting patterns of sedentary behaviour known to increase cardiometabolic risk. A number of practical factors, including the incorporation of exercises that target major muscle groups, are osteogenic in nature, and challenge both balance and the stabilising muscles, are likely to influence the effectiveness of minimal-dose RT approaches for improving muscle function and bone health and, in turn, for improving independence and quality-of-life. The combination of resistance and aerobic exercise has been shown to have the greatest benefits for improving overall health and reducing risk of all-cause mortality, suggesting more work is needed to determine the most effective ways to integrate both minimal-dose resistance and aerobic exercise approaches into an exercise regime. Although further evidence is needed, minimal-dose approaches to resistance exercise have the potential to substantially improve population health in a time-efficient, feasible, and scalable manner. Future research in this area should aim to address the following questions in particular:

- What is the minimum effective dose and potential dose–response effects for improvements in strength and power with minimal-dose RT (i.e. low-volume, high-load RT or resistance ‘exercise snacking’)?
- Are improvements in strength with minimal-dose RT approaches attributed to greater opportunities to ‘practise the test’ during the exercise programme, or do strength benefits transfer to non-specific activities?
- Do minimal-dose RT approaches lead to meaningful improvements in functional capacity in older adults (e.g. walking speed, sit-to-stand performance)?

- Can the application of non-exercise strategies (e.g. tel-ehealth) improve the feasibility of, and adherence to, minimal-dose RT when delivered remotely?
- What is the influence of additional aerobic exercise on the dose–response relationship for the efficacy of low-volume, high-load and resistance ‘exercise snacking’ protocols for improving strength?

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Declarations

Conflict of interest Jackson Fyfe, D. Lee Hamilton, and Robin Daly declare that they have no conflicts of interest relevant to the content of this review.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Availability of data and material Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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